

Analysis of Salary Data

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Agenda

Findings of linear regression modeling with tech salary data

- Data Description
- Regression Results
- Interpretation and Next Steps

Data Description - Before Cleaning

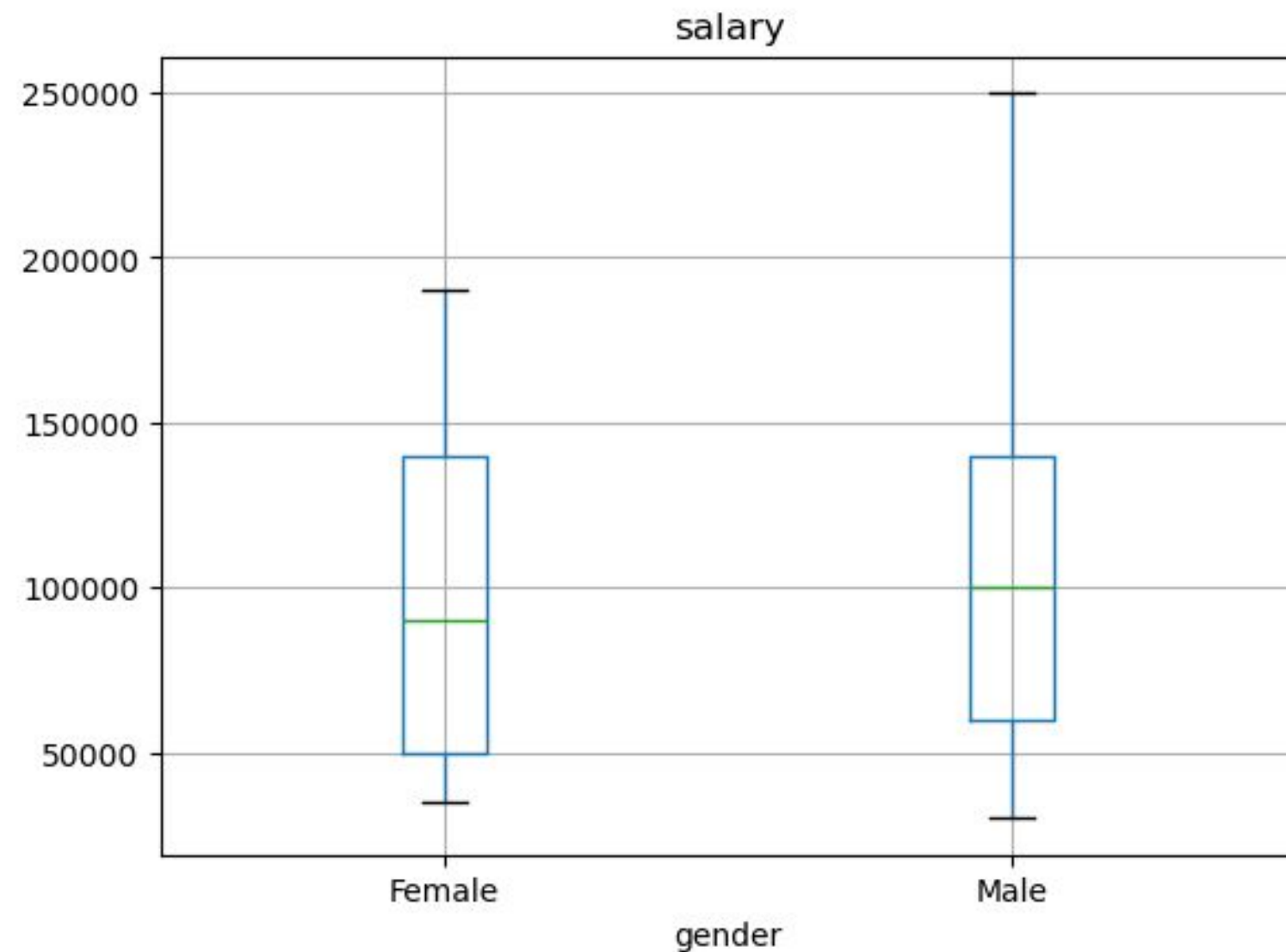
Before cleaning there are 375 employees in the dataset. The salaries in the dataset ranged from \$350 to \$250,000. The dataset contained the following columns and data types.

Column Name	Data Type
Age	float
Gender	string
Education Level	string
Job Title	string
Years of Experience	float
Salary	float

Data Description - Data Cleaning

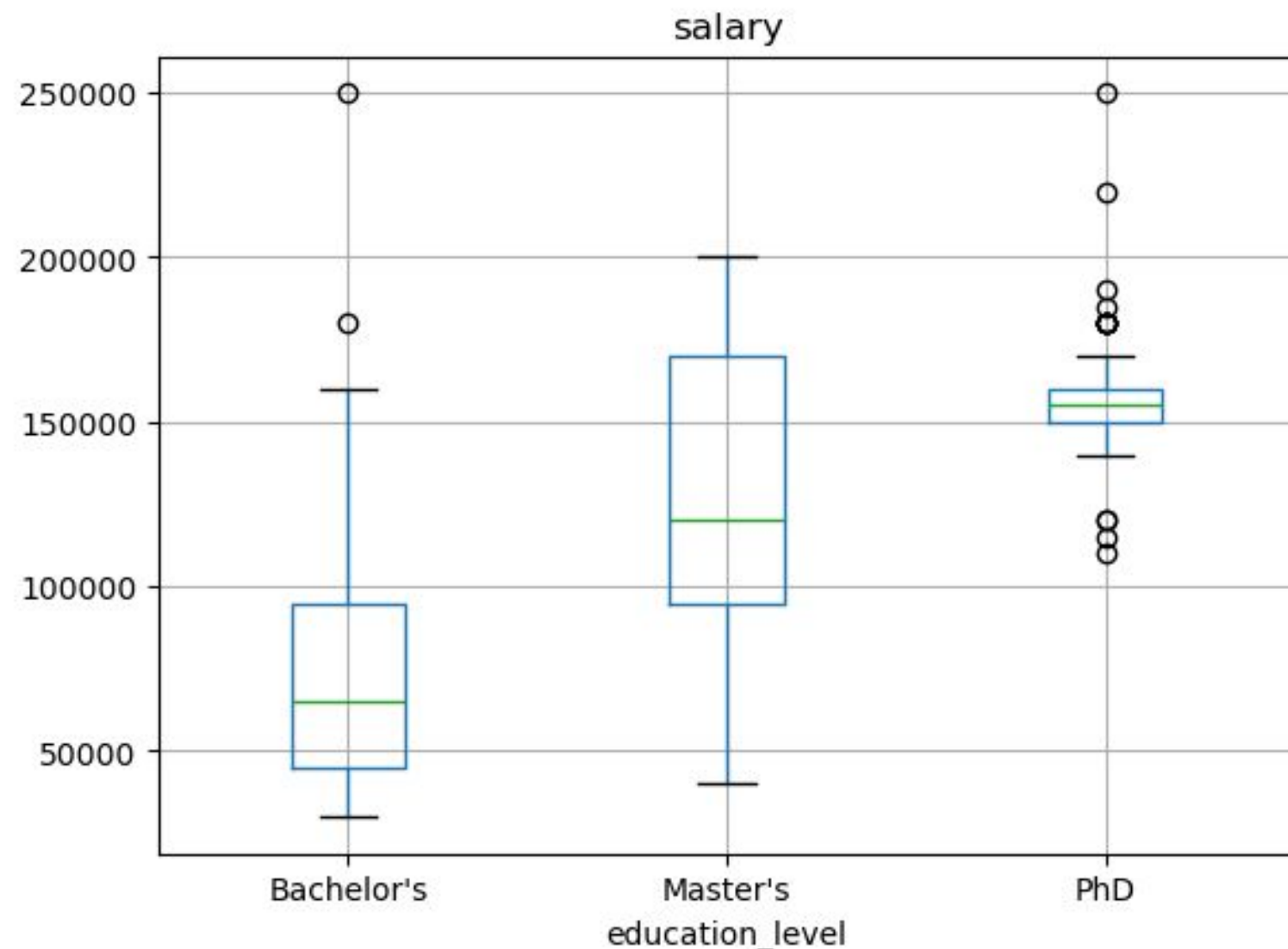
- Since the \$350 salary value was so far out of range and likely an error given the context of the dataset, the value was dropped.
- Each column in the dataset contained two missing values. The missing values were all in the same two blank rows, which were dropped.
- 372 employees remained after cleaning.
- The column names were changed to be lowercase with underscores separating the words, in keeping with Python best practices.

Data Description - Gender



- The average salary for men is around 8% higher than for women.
- The median salary for men is around 11% higher than it is for women.

Data Description - Education Level



- The median salary for a master's degree is about 85% higher than for a bachelor's degree.
- The median salary for a doctorate is about 138% higher than for a bachelor's degree, and about 29% higher than for a master's degree.

Data Description - Years of Experience



There is a strong positive correlation (0.93) between salary and years of experience.

Regression Results - Initial Inputs

Regression Model: Ordinary Least Squares

Dependent variable: Salary

Predictors: Years of Experience, Job Title, Gender, Level of Education

OLS Regression Results			
=====			
Dep. Variable:	salary	R-squared:	0.917
Model:	OLS	Adj. R-squared:	0.914
Method:	Least Squares	F-statistic:	396.3
Date:	Thu, 24 Sep 2026	Prob (F-statistic):	7.55e-188
Time:	11:59:05	Log-Likelihood:	-4075.5
No. Observations:	372	AIC:	8173.
Df Residuals:	361	BIC:	8216.
Df Model:	10		
Covariance Type:	nonrobust		

Regression Results - Initial Accuracy

- R-squared value of 0.917
- Selected predictors explain about 92% of salary variability.
- F-statistic of 396.3 and p-value ≈ 0 tell us the model as a whole is statistically significant.

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Regression Results - Initial Variable Significance

The following variables were insignificant (p-value > 0.05):

- is_junior: 0.089
- is_analyst: 0.846
- is_engineer: 0.501

	coef	std err	t	P> t	[0.025	0.975]
const	3.2e+04	2458.521	13.014	0.000	2.72e+04	3.68e+04
years_of_experience	4928.4788	195.393	25.223	0.000	4544.227	5312.730
is_director	2.554e+04	3496.600	7.305	0.000	1.87e+04	3.24e+04
is_junior	-4156.5912	2433.955	-1.708	0.089	-8943.104	629.921
is_senior	1.302e+04	1955.098	6.659	0.000	9174.774	1.69e+04
is_manager	4503.6322	1938.874	2.323	0.021	690.725	8316.539
is_analyst	406.1927	2089.609	0.194	0.846	-3703.142	4515.527
is_engineer	2371.1630	3522.702	0.673	0.501	-4556.433	9298.759
gender_flag	7676.6582	1494.609	5.136	0.000	4737.424	1.06e+04
Master's	1.346e+04	2031.258	6.628	0.000	9468.287	1.75e+04
PhD	2.338e+04	2771.286	8.436	0.000	1.79e+04	2.88e+04

Regression Results - Optimized Inputs

Removed: is_junior, is_analyst, is_engineer

OLS Regression Results			
=====			
Dep. Variable:	salary	R-squared:	0.916
Model:	OLS	Adj. R-squared:	0.914
Method:	Least Squares	F-statistic:	564.9
Date:	Thu, 24 Sep 2026	Prob (F-statistic):	3.42e-191
Time:	11:59:05	Log-Likelihood:	-4077.2
No. Observations:	372	AIC:	8170.
Df Residuals:	364	BIC:	8202.
Df Model:	7		
Covariance Type:	nonrobust		

Regression Results - Optimized Accuracy

- Adjusted R-squared remains 0.914
- Selected predictors still explain about 92% of salary variability.
- F-statistic p-value still shows the model is statistically significant.

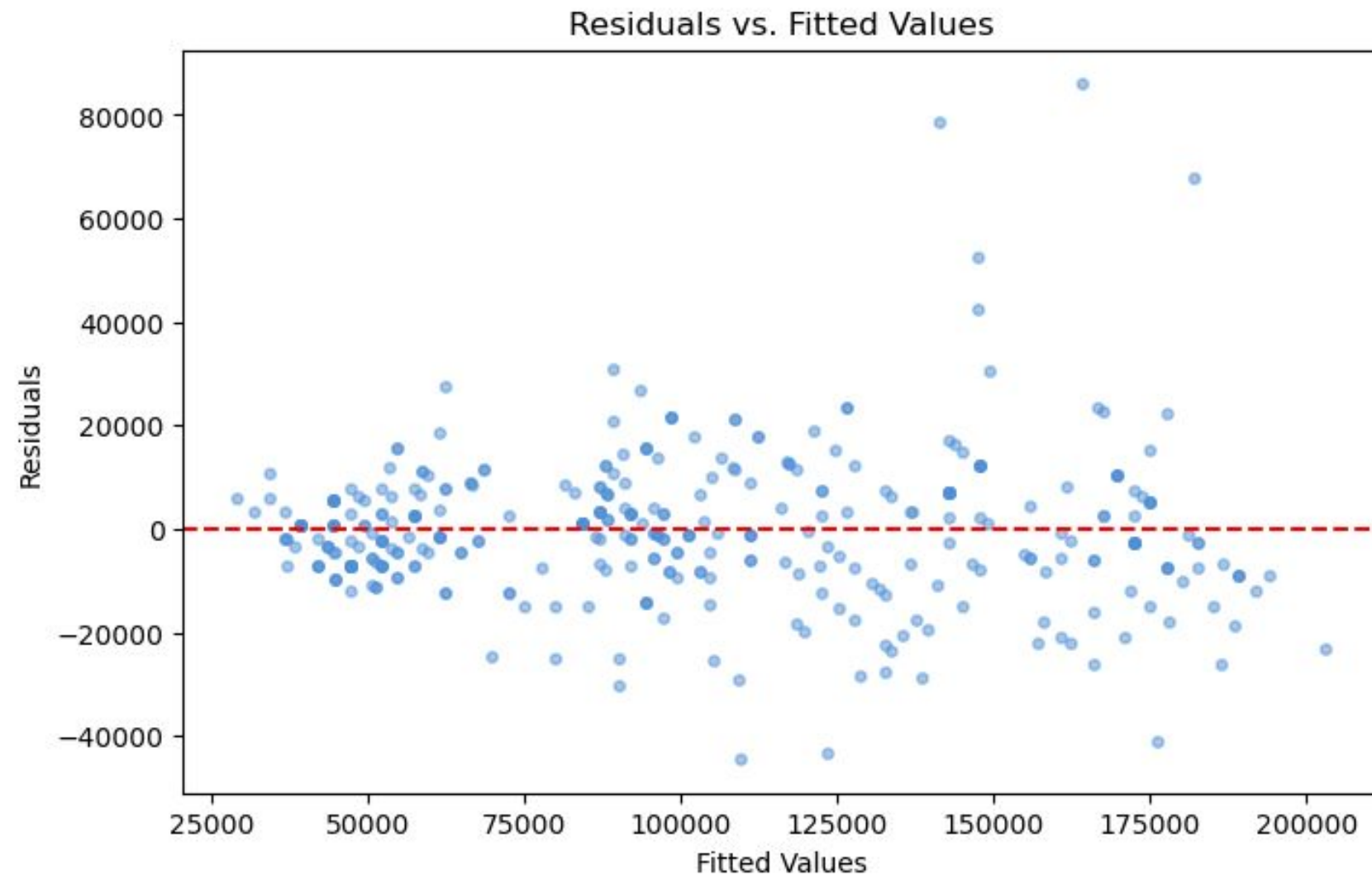
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Regression Results - Optimized Variable Significance

All remaining variables are statistically significant.

	coef	std err	t	P> t	[0.025	0.975]
const	2.907e+04	1660.594	17.505	0.000	2.58e+04	3.23e+04
years_of_experience	5081.2555	175.297	28.987	0.000	4736.533	5425.978
is_director	2.505e+04	3461.875	7.237	0.000	1.82e+04	3.19e+04
is_senior	1.435e+04	1799.292	7.973	0.000	1.08e+04	1.79e+04
is_manager	4174.9452	1801.474	2.318	0.021	632.342	7717.549
gender_flag	7848.8370	1466.075	5.354	0.000	4965.796	1.07e+04
Master's	1.397e+04	2008.268	6.958	0.000	1e+04	1.79e+04
PhD	2.314e+04	2728.306	8.480	0.000	1.78e+04	2.85e+04

Regression Results - Final Model Fit



R-squared: 91.6% salary variability accounted for

RMSE: ~\$13,924

- 14% of the mean salary
- 71% lower than the \$48,023 stddev

Less reliable at higher salary ranges

- residual spread increases as predicted salary increases

Interpretation and Next Steps

The data shows that higher education and more experience are associated with higher salaries.

We can be 95% confident that salary increases by between \$4,736.53 and \$5,425.98 for each additional year of experience, holding all else constant.

We can be 95% confident that salary is between \$17,771.82 and \$28,502.26 higher for someone with a PhD than for someone with a bachelor's degree, holding all else constant.

Since the model's residuals show greater spread at higher salary ranges, further exploration and modeling should be done to determine which factors impact salary at higher ranges.